



兰州工业学院

实 验 报 告

(2023—2024 学年第 1 学期)

课程名称 《网络操作系统》

专业班级 专升本电信 23

学 号 ZB2305010116

姓 名 姜 勇

实验项目	实验四 Linux 文本处理实验		
实验时间	2023 年 10 月 27 日	实验地点	信息与通信工程基础实验室 (EDA 技术实验室)

一、实验内容

使用文本处理器进行文本的编辑、切片、比较命令等；使用正则表达式来对 Linux 中文件进行相关的处理和操作；通过对文本处理器的配置的修改和自定义，打造属于自己的文本编辑器。

二、实验器材（设备、元器件）

电脑、VMware Workstation Pro 虚拟机软件包、Centos 镜像包。

三、实验步骤及结果分析

第一部分

请尽量使用键盘完成全部操作

（一）前期准备

（1）复制 /etc/manpath.config 文件到当前目录，使用 vim 打开本目录下的 manpath.config 文件，在 vi 中设置一下行号和查找结果高亮。

设置行号: *set nu*

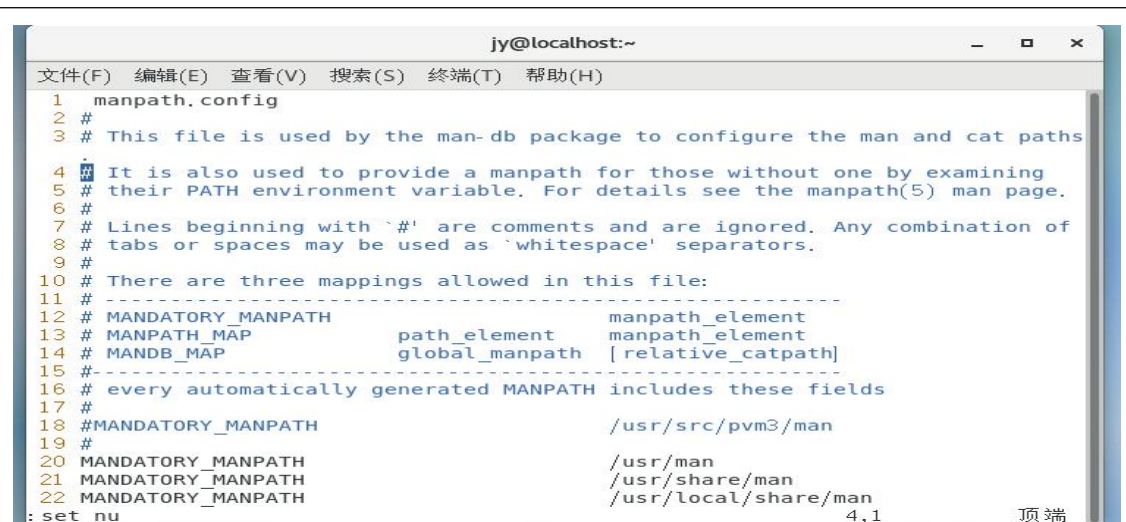
设置搜索高亮: *set hlsearch*

图 1



图 1 使用 cp 命令将/etc/manpath.config 复制到了当前目录,使用 ls 命令查看是否复制成功

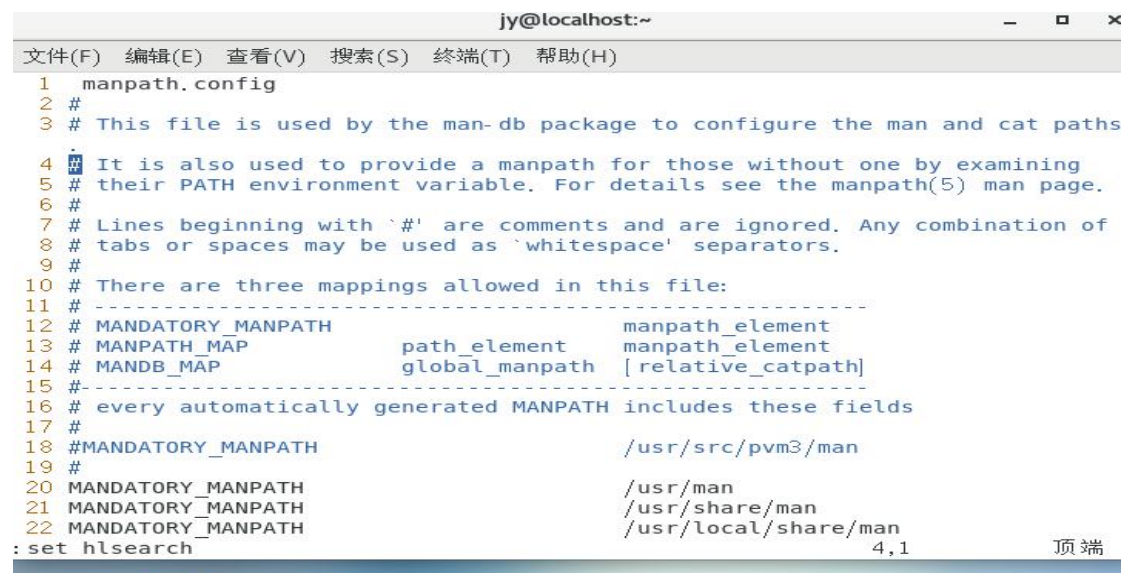
图 2



```
1 manpath.config
2 #
3 # This file is used by the man-db package to configure the man and cat paths
4 #
5 # It is also used to provide a manpath for those without one by examining
6 # their PATH environment variable. For details see the manpath(5) man page.
7 #
8 # Lines beginning with '#' are comments and are ignored. Any combination of
9 # tabs or spaces may be used as 'whitespace' separators.
10 #
11 # There are three mappings allowed in this file:
12 # -----
13 # MANDATORY_MANPATH      path_element      manpath_element
14 # MANPATH_MAP             global_manpath    [relative_catpath]
15 # -----
16 # every automatically generated MANPATH includes these fields
17 #
18 #MANDATORY_MANPATH      /usr/src/pvm3/man
19 #
20 MANDATORY_MANPATH      /usr/man
21 MANDATORY_MANPATH      /usr/share/man
22 MANDATORY_MANPATH      /usr/local/share/man
: set nu                                     4,1      顶端
```

图 2 在 vim 末行模式使用 set nu 代码为文件设置了行号

图 3



```
1 manpath.config
2 #
3 # This file is used by the man-db package to configure the man and cat paths
4 #
5 # It is also used to provide a manpath for those without one by examining
6 # their PATH environment variable. For details see the manpath(5) man page.
7 #
8 # Lines beginning with '#' are comments and are ignored. Any combination of
9 # tabs or spaces may be used as 'whitespace' separators.
10 #
11 # There are three mappings allowed in this file:
12 # -----
13 # MANDATORY_MANPATH      path_element      manpath_element
14 # MANPATH_MAP             global_manpath    [relative_catpath]
15 # -----
16 # every automatically generated MANPATH includes these fields
17 #
18 #MANDATORY_MANPATH      /usr/src/pvm3/man
19 #
20 MANDATORY_MANPATH      /usr/man
21 MANDATORY_MANPATH      /usr/share/man
22 MANDATORY_MANPATH      /usr/local/share/man
: set hlsearch                             4,1      顶端
```

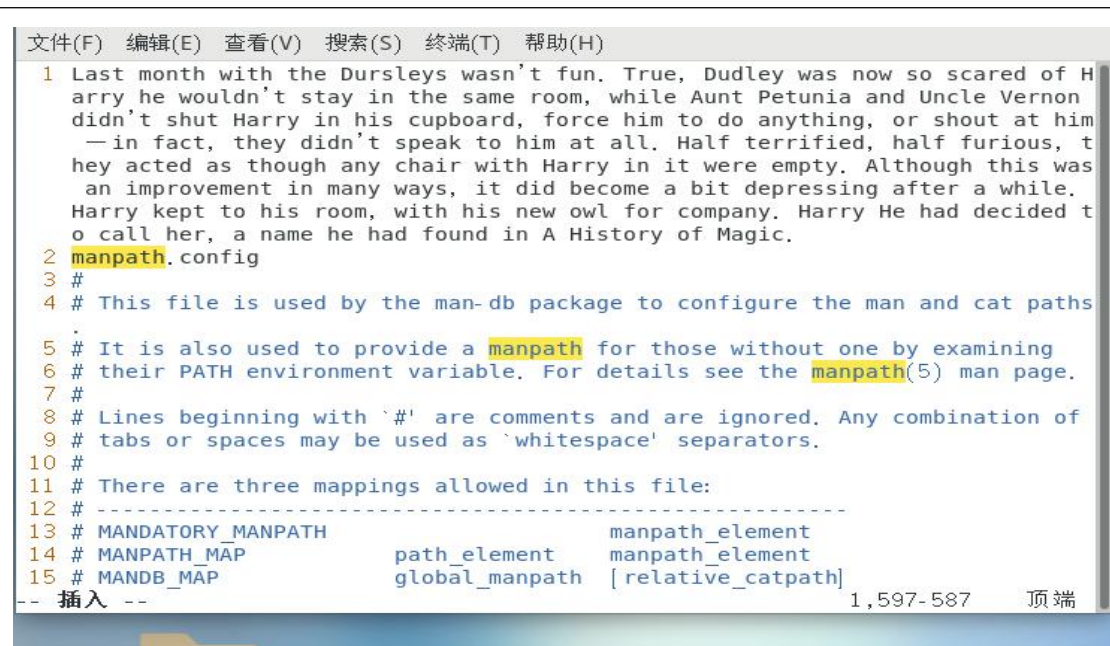
图 3 在 vim 末行模式使用 set hlsearch 代码为文件设置了查找的结果高亮显示

(二) 完成对应的操作

(1) 进入编辑模式，并在开头输入相关文字为：

Last month with the Dursleys wasn't fun. True, Dudley was now so scared of Harry he wouldn't stay in the same room, while Aunt Petunia and Uncle Vernon didn't shut Harry in his cupboard, force him to do anything, or shout at him — in fact, they didn't speak to him at all. Half terrified, half furious, they acted as though any chair with Harry in it were empty. Although this was an improvement in many ways, it did become a bit depressing after a while. Harry kept to his room, with his new owl for company. Harry He had decided to call her, a name he had found in A History of Magic.

图 4



```
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
1 Last month with the Dursleys wasn't fun. True, Dudley was now so scared of H
arry he wouldn't stay in the same room, while Aunt Petunia and Uncle Vernon
didn't shut Harry in his cupboard, force him to do anything, or shout at him
—in fact, they didn't speak to him at all. Half terrified, half furious, t
hey acted as though any chair with Harry in it were empty. Although this was
an improvement in many ways, it did become a bit depressing after a while.
Harry kept to his room, with his new owl for company. Harry He had decided t
o call her, a name he had found in A History of Magic.
2 manpath.config
3 #
4 # This file is used by the man-db package to configure the man and cat paths
.
5 # It is also used to provide a manpath for those without one by examining
6 # their PATH environment variable. For details see the manpath(5) man page.
7 #
8 # Lines beginning with '#' are comments and are ignored. Any combination of
9 # tabs or spaces may be used as 'whitespace' separators.
10 #
11 # There are three mappings allowed in this file:
12 # -----
13 # MANDATORY_MANPATH          manpath_element
14 # MANPATH_MAP                path_element  manpath_element
15 # MANDB_MAP                  global_manpath [relative_catpath]
-- 插入 -- 1,597-587 顶端
```

图 4 在 vim 命令模式下按 i 进入插入模式粘贴了这段文字
(2) 在命令模式下，将 (1) 的文字复制到文件的最后。

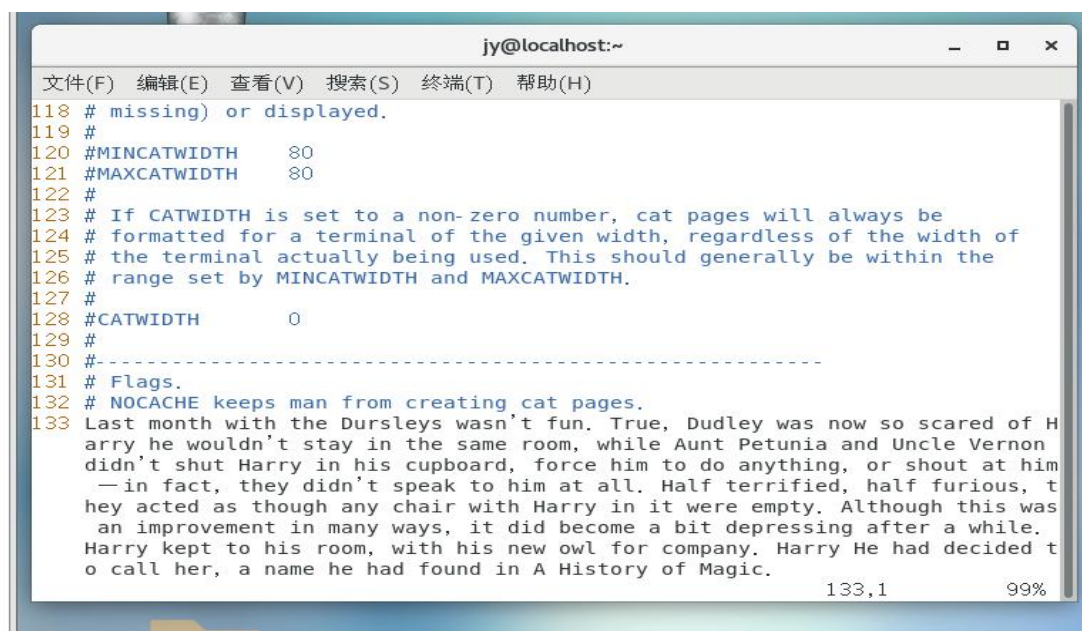
跳到第 n 行: nG

复制 n 行: nyy

粘贴: p

撤销: u

图 5



```
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
118 # missing) or displayed.
119 #
120 #MINCATWIDTH      80
121 #MAXCATWIDTH      80
122 #
123 # If CATWIDTH is set to a non-zero number, cat pages will always be
124 # formatted for a terminal of the given width, regardless of the width of
125 # the terminal actually being used. This should generally be within the
126 # range set by MINCATWIDTH and MAXCATWIDTH.
127 #
128 #CATWIDTH          0
129 #
130 # -----
131 # Flags.
132 # NOCACHE keeps man from creating cat pages.
133 Last month with the Dursleys wasn't fun. True, Dudley was now so scared of H
arry he wouldn't stay in the same room, while Aunt Petunia and Uncle Vernon
didn't shut Harry in his cupboard, force him to do anything, or shout at him
—in fact, they didn't speak to him at all. Half terrified, half furious, t
hey acted as though any chair with Harry in it were empty. Although this was
an improvement in many ways, it did become a bit depressing after a while.
Harry kept to his room, with his new owl for company. Harry He had decided t
o call her, a name he had found in A History of Magic.
133,1 99%
```

图 5 在命令模式下使用 1G 命令跳到了第一行，使用 yy 命令复制了这段文字，再使用 G 命令跳到了最后一行，使用 p 粘贴

(3) 在命令模式下，使用查找命令查找 DB_MAP，并将找到的行所包含的 man 修改为 MAN

搜索字符串: /<string>

将第 n~m 行的 string1 替换为 string2: :n,ms/string1/string2/gc

图 6

```
64 #
65 #          *MANPATH*      ->      *CATPATH*
66 #
67 MANDB_MAP      /usr/man      /var/cache/man/fsstnd
68 MANDB_MAP      /usr/share/man /var/cache/man
69 MANDB_MAP      /usr/local/man /var/cache/man/oldlocal
70 MANDB_MAP      /usr/local/share/man /var/cache/man/local
71 MANDB_MAP      /usr/X11R6/man /var/cache/man/X11R6
72 MANDB_MAP      /opt/man      /var/cache/man/opt
73 MANDB_MAP      /snap/man     /var/cache/man/snap
74 #
75 #-----
76 # Program definitions. These are commented out by default as the value
77 # of the definition is already the default. To change: uncomment a
78 # definition and modify it.
79 #
80 #DEFINE      pager      pager
81 #DEFINE      cat      cat
                                     59,1      52%
```

图 6 在命令模式下使用/DB_MAP 查找了 DB_MAP

图 7

```
hey acted as though any chair with Harry in it were empty. Although this was
an improvement in MANY ways, it did become a bit depressing after a while.
Harry kept to his room, with his new owl for company. Harry He had decided t
o call her, a name he had found in A History of Magic.
2 MANpath.config
3 #
4 # This file is used by the man-db package to configure the man and cat paths
5 #
6 # It is also used to provide a manpath for those without one by examining
7 # their PATH environment variable. For details see the manpath(5) man page.
8 #
9 # Lines beginning with '#' are comments and are ignored. Any combination of
10 # tabs or spaces may be used as 'whitespace' separators.
11 #
12 # There are three mappings allowed in this file:
13 # -----
14 # MANDATORY_MANPATH      path_element      manpath_element
15 # MANPATH_MAP            global_manpath     [relative_catpath]
替换为 MAN (y/n/a/q/l/^E/^Y)? 4,28      顶端
```

图 7 在末行模式下使用:n, ms/man/MAN/gc 命令替换了所有 “man” 为 “MAN”，
每一步输入 y 来确认

(4) 在命令模式下，将全文中的 Harry 替换为 Potter

将全文的 string1 替换为 string2: :1,\$s/string1/string2/gc

图 8

```

文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
1 Last month with the Dursleys wasn't fun. True, Dudley was now so scared of
  arry he wouldn't stay in the same room, while Aunt Petunia and Uncle Vernon
  didn't shut Harry in his cupboard, force him to do anything, or shout at him
  —in fact, they didn't speak to him at all. Half terrified, half furious, t
  hey acted as though any chair with Harry in it were empty. Although this was
  an improvement in MANY ways, it did become a bit depressing after a while.
  Harry kept to his room, with his new owl for company. Harry He had decided t
  o call her, a name he had found in A History of Magic.
2 MANpath.config
3 #
4 # This file is used by the MAN-db package to configure the MAN and cat paths
5 #
6 # It is also used to provide a MANpath for those without one by examining
7 # their PATH environment variable. For details see the MANpath(5) MAN page.
8 #
9 # Lines beginning with '#' are comments and are ignored. Any combination of
10 # tabs or spaces may be used as 'whitespace' separators.
11 # There are three mappings allowed in this file:
12 # -----
13 # MANDATORY_MANPATH      path_element      MANpath_element
14 # MANPATH_MAP            global_MANpath    [relative_catpath]
15 # MANDB_MAP
替换为 Potter (y/n/a/q/l/^E/^Y)? 1,78-76 顶端

```

图 8 在命令模式下，使用:1,\$s/Harry/Potter/gc 将全文中的 Harry 替换成了 Potter

(5) 在命令模式下，复制文章最后的部分中标黄的句子 Although this was an improvement in many ways, 复制后将这句话进行删除

- 进入选择模式: v
- 向后选择 n 个字符: n<space>
- 在选择模式下复制: y
- 删除字符: x
- 删除 n 个字符: nx

图 9

```

jy@localhost:~
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
119 #
120 #MINCATWIDTH      80
121 #MAXCATWIDTH      80
122 #
123 # If CATWIDTH is set to a non-zero number, cat pages will always be
124 # formatted for a terminal of the given width, regardless of the wi
125 # the terminal actually being used. This should generally be within
126 # range set by MINCATWIDTH and MAXCATWIDTH.
127 #
128 #CATWIDTH          0
129 #
130 #-----
131 # Flags.
132 # NOCACHE keeps MAN from creating cat pages.
133 Last month with the Dursleys wasn't fun. True, Dudley was now so sc
  otter he wouldn't stay in the same room, while Aunt Petunia and Unc
  didn't shut Potter in his cupboard, force him to do anything, or s
  im —in fact, they didn't speak to him at all. Half terrified, half
  they acted as though any chair with Potter in it were empty. Altho
  was an improvement in MANY ways, it did become a bit depressing aft
  e. Potter kept to his room, with his new owl for company. Potter He
  ded to call her, a name he had found in A History of Magi
134 Although this was an improvement in MANY ways, #NOCACHE
134,48

```

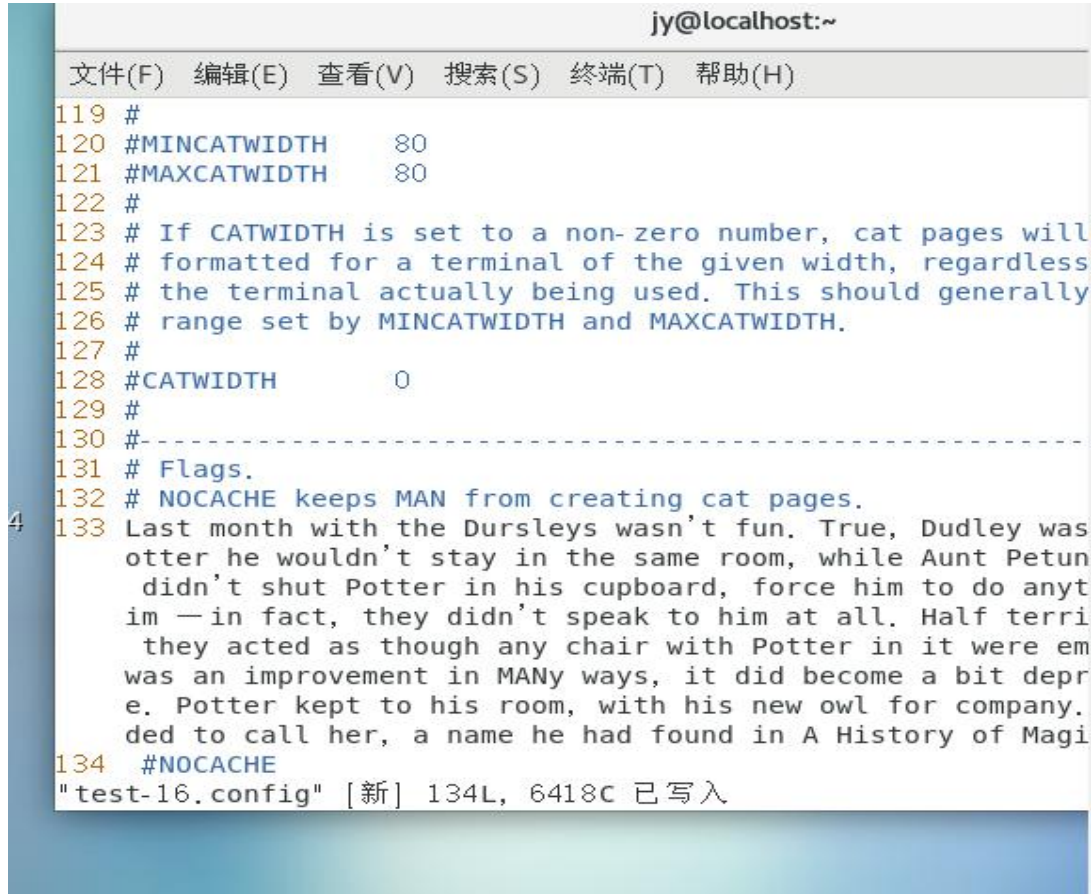
图 9 在命令模式下，按 v 进入选择模式，选择了标黄的句子，使用 y 复制，然

后使用 **x** 将这句话删除

(6) 保存文件名为: **test-<学号>.config**

保存文件名为 **fileName**: **:w fileName**

图 10



```
jy@localhost:~
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
119 #
120 #MINCATWIDTH      80
121 #MAXCATWIDTH      80
122 #
123 # If CATWIDTH is set to a non-zero number, cat pages will
124 # formatted for a terminal of the given width, regardless
125 # the terminal actually being used. This should generally
126 # range set by MINCATWIDTH and MAXCATWIDTH.
127 #
128 #CATWIDTH          0
129 #
130 #-----
131 # Flags.
132 # NOCACHE keeps MAN from creating cat pages.
4 133 Last month with the Dursleys wasn't fun. True, Dudley was
    otter he wouldn't stay in the same room, while Aunt Petun
    didn't shut Potter in his cupboard, force him to do anyt
    im — in fact, they didn't speak to him at all. Half terri
    they acted as though any chair with Potter in it were em
    was an improvement in MANY ways, it did become a bit depr
    e. Potter kept to his room, with his new owl for company.
    ded to call her, a name he had found in A History of Magi
134 #NOCACHE
"test-16.config" [新] 134L, 6418C 已写入
```

图 10 在末行模式下使用 **w test-16.config** 将文件另存

第二部分

以下命令的使用中注意转义符“\”的使用

使用正则表达式的命令最常见的就是 **/** 和 **?** 命令。其格式如下：

/正则表达式

?正则表达式

另一个很有用的命令就是 **:s**（替换）命令，将第一个 **//** 之间的正则表达式替换成第二个 **//** 之间的字符串。

:s/正则表达式/替换字符串/选项

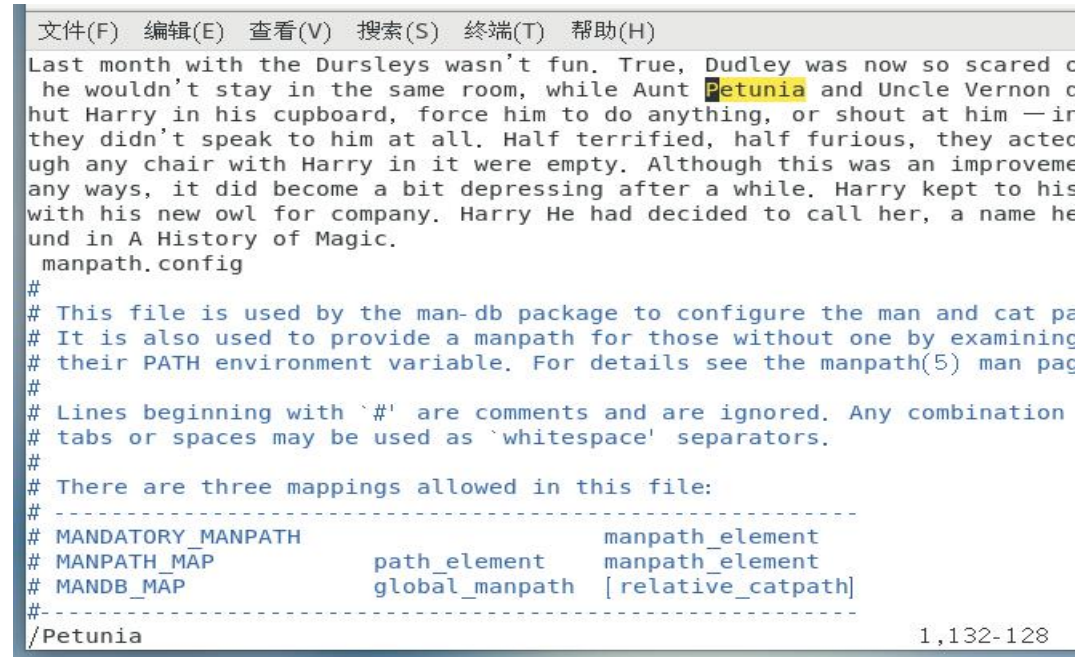
通过使用 **grep** 和 **vim** 来检索上次实验内容，并完成如下操作。

1. 基本匹配

正则表达式其实就是在执行搜索时的格式，它由一些字母和数字组合而成。例如：一个正则表达式 `the`，它表示一个规则：由字母 `t` 开始，接着是 `h`，再接着是 `e`。

```
/Harry
grep Harry <filename>
```

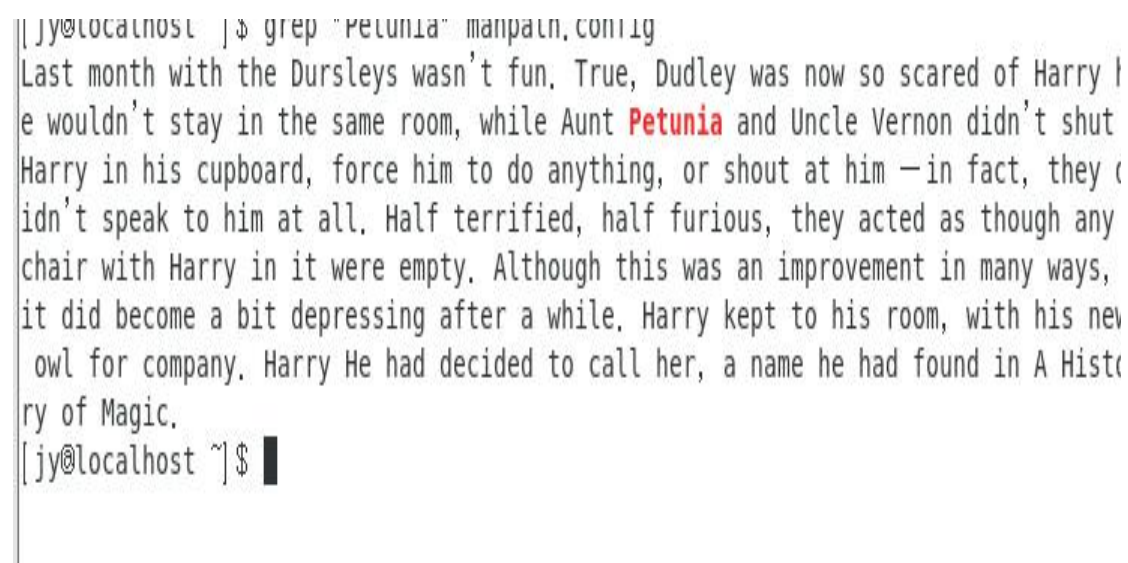
图 11



```
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
Last month with the Dursleys wasn't fun. True, Dudley was now so scared c
he wouldn't stay in the same room, while Aunt Petunia and Uncle Vernon c
hut Harry in his cupboard, force him to do anything, or shout at him —ir
they didn't speak to him at all. Half terrified, half furious, they acted
ugh any chair with Harry in it were empty. Although this was an improveme
any ways, it did become a bit depressing after a while. Harry kept to his
with his new owl for company. Harry He had decided to call her, a name he
und in A History of Magic.
manpath.config
#
# This file is used by the man-db package to configure the man and cat pa
# It is also used to provide a manpath for those without one by examining
# their PATH environment variable. For details see the manpath(5) man pag
#
# Lines beginning with '#' are comments and are ignored. Any combination
# tabs or spaces may be used as 'whitespace' separators.
#
# There are three mappings allowed in this file:
# -----
# MANDATORY_MANPATH          manpath_element
# MANPATH_MAP                path_element    manpath_element
# MANDB_MAP                  global_manpath  [relative_catpath]
# -----
/Petunia 1,132-128
```

图 11 在命令模式在 `/Petunia` 查找了查找文档中所有包含 “Harry” 的单词

图 12



```
[jy@localhost ~]$ grep "Petunia" manpath.config
Last month with the Dursleys wasn't fun. True, Dudley was now so scared of Harry l
e wouldn't stay in the same room, while Aunt Petunia and Uncle Vernon didn't shut
Harry in his cupboard, force him to do anything, or shout at him —in fact, they c
idn't speak to him at all. Half terrified, half furious, they acted as though any
chair with Harry in it were empty. Although this was an improvement in many ways,
it did become a bit depressing after a while. Harry kept to his room, with his new
owl for company. Harry He had decided to call her, a name he had found in A Histo
ry of Magic.
[jy@localhost ~]$
```

图 12 在终端使用 `grep` 命令来在文件 `manpath.config` 中查找包含 Harry 的行

2.1 点运算符 .

.是元字符中最简单的例子。 .匹配任意单个字符，但不匹配换行符。 例如，表达式`.ar` 匹配一个任意字符后面跟着是 `a` 和 `r` 的字符串。

```
/.ry  
grep .ry <filename>
```

图 13

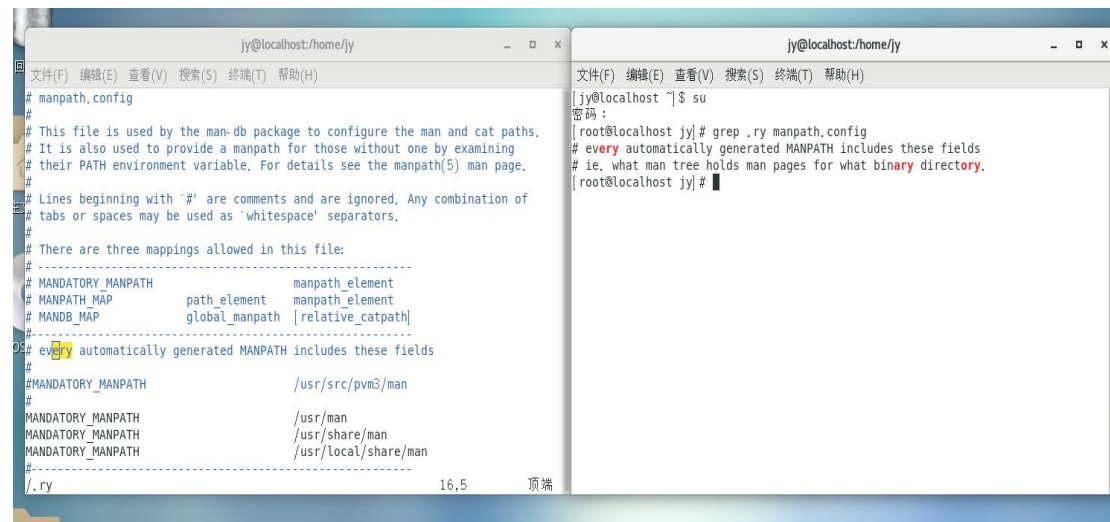


图 13 在命令模式下，输入 `/.ry`，即可查找文档中所有包含 “ry” 字符序列的单词，在终端使用 `grep` 命令来在文件 `manpath.config` 中查找以“`.ry`”结尾的字符串

2.2 字符集

字符集也叫做字符类。方括号用来指定一个字符集。在方括号中使用连字符来指定字符集的范围。在方括号中的字符集不关心顺序。例如，表达式`[Tt]he` 匹配 `the` 和 `The`。

```
/[ar]ry  
grep [ar]ry <filename>
```

图 14

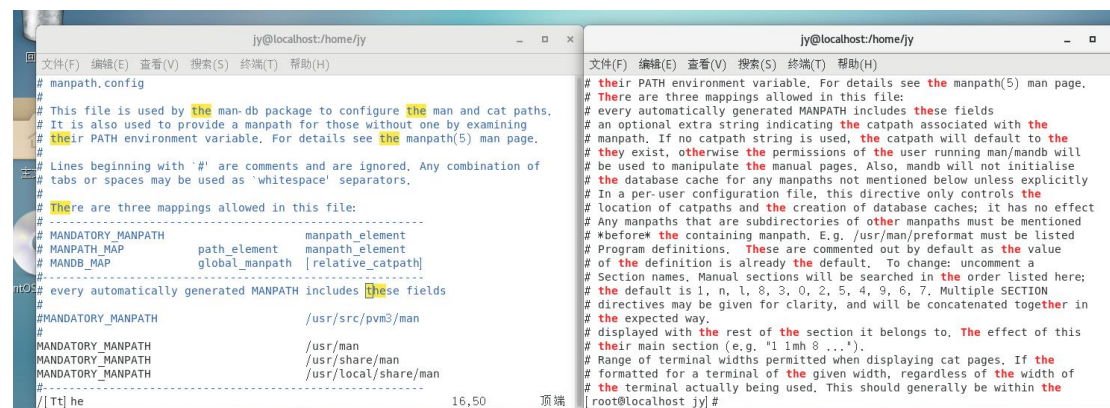


图 14 在命令模式下，输入 `/[ar]ry` 可以查找当前文档中以 “array” 或

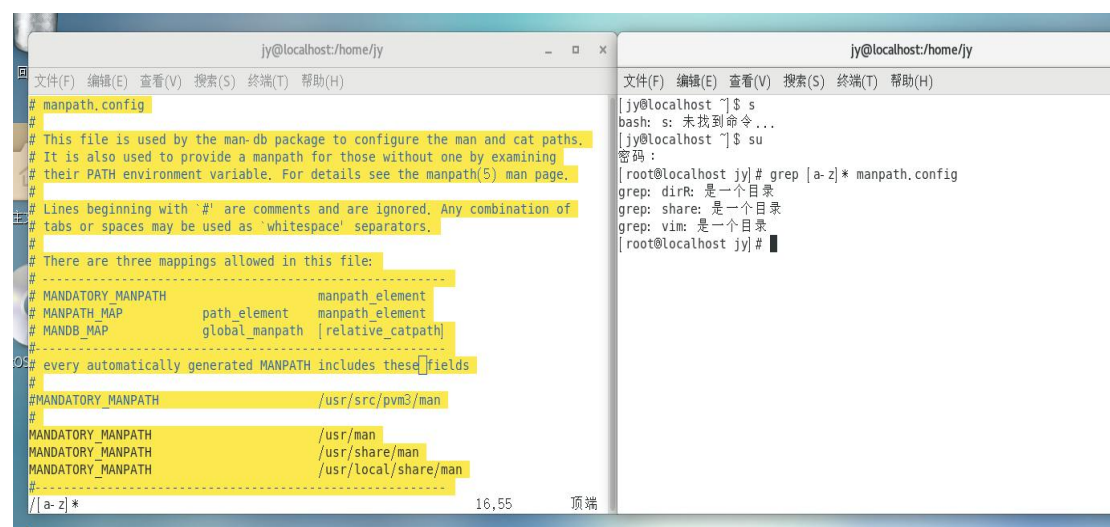
“rry” 结尾的字符串，在终端使用 `grep` 命令来在文件 `manpath.config` 中查找以 “arry” 或 “rry” 结尾的字符串

2.3 重复次数

后面跟着元字符 `+`、`*` 或 `?` 的，用来指定匹配子模式的次数。这些元字符在不同的情况下有着不同的意思。

```
/[a-z]*  
grep [a-z]* <filename>  
/[r]?y  
grep [r]?y <filename>
```

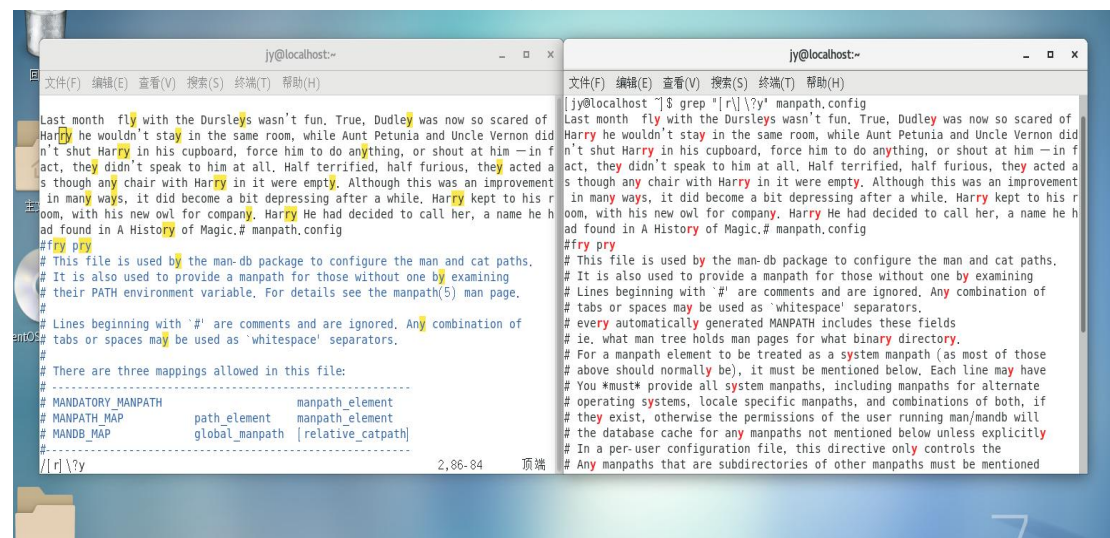
图 15



```
jy@localhost/home/jy  
# manpath.config  
# This file is used by the man-db package to configure the man and cat paths.  
# It is also used to provide a manpath for those without one by examining  
# their PATH environment variable. For details see the manpath(5) man page.  
# Lines beginning with '#' are comments and are ignored. Any combination of  
# tabs or spaces may be used as 'whitespace' separators.  
# There are three mappings allowed in this file:  
#-----  
# MANDATORY_MANPATH      manpath_element  
# MANPATH_MAP             path_element    manpath_element  
# MANDB_MAP               global_manpath  [relative_catpath]  
#-----  
# every automatically generated MANPATH includes these fields  
#-----  
# MANDATORY_MANPATH      /usr/src/pvm3/man  
#-----  
# MANDATORY_MANPATH      /usr/man  
# MANDATORY_MANPATH      /usr/share/man  
# MANDATORY_MANPATH      /usr/local/share/man  
#-----  
/[a-z]*
```

图 15 在命令模式下输入 `/[a-z]*` 查找当前文档中包含小写字母的连续字符串，在终端使用 `grep` 命令来查找文件中包含小写字母的连续字符串

图 16



```
jy@localhost:~  
# manpath.config  
# This file is used by the man-db package to configure the man and cat paths.  
# It is also used to provide a manpath for those without one by examining  
# their PATH environment variable. For details see the manpath(5) man page.  
# Lines beginning with '#' are comments and are ignored. Any combination of  
# tabs or spaces may be used as 'whitespace' separators.  
# There are three mappings allowed in this file:  
#-----  
# MANDATORY_MANPATH      manpath_element  
# MANPATH_MAP             path_element    manpath_element  
# MANDB_MAP               global_manpath  [relative_catpath]  
#-----  
# every automatically generated MANPATH includes these fields  
#-----  
# MANDATORY_MANPATH      /usr/src/pvm3/man  
#-----  
# MANDATORY_MANPATH      /usr/man  
# MANDATORY_MANPATH      /usr/share/man  
# MANDATORY_MANPATH      /usr/local/share/man  
#-----  
/[r]?y
```

图 16 在命令模式下输入 `/[r]?y` 可以查找当前文档中以 “y” 结尾，

前面可能有一个“r”的字符串，在终端使用 `grep` 命令来查找文件中以“y”结尾，前面可能有一个“r”的字符串

2.4 {} 号

在正则表达式中 `{}` 是一个量词，常用来限定一个或一组字符可以重复出现的次数。例如，表达式 `[0-9]{2,3}` 匹配最少 2 位最多 3 位 0~9 的数字。

```
/Ha[r]{2,3}y
```

```
grep Ha[r]{2,3}y <filename>
```

图 17

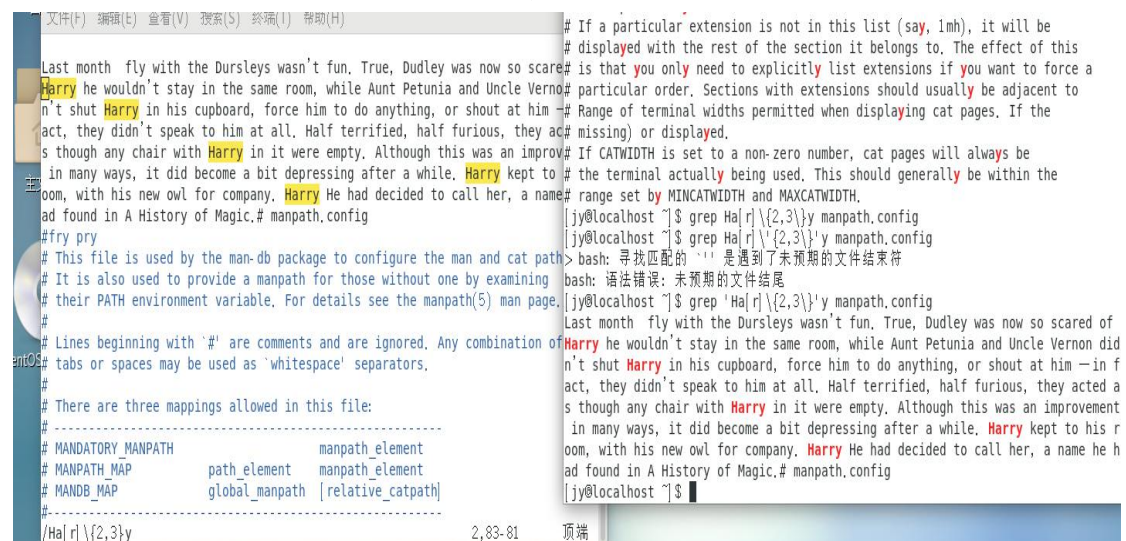
A screenshot of a terminal window. The top part shows the output of a `grep` command: `grep 'Ha[r]{2,3}y' manpath.config`. The output is a list of lines from the `manpath.config` file, including comments and configuration settings. The bottom part shows the command prompt `bash: 语法错误: 未预期的文件结尾` (syntax error: unexpected end of file) and the command `grep 'Ha[r]{2,3}y' manpath.config` being entered again.

图 17 在命令模式下输入 `/Ha[r]{2,3}y` 可以查找当前文档中以“Ha”开头，接着是 2 到 3 个连续的“r”，最后以“y”结尾的字符串，在终端使用 `grep` 命令来查找文档中以“Ha”开头，接着是 2 到 3 个连续的“r”，最后以“y”结尾的字符串

2.5 (...) 特征标群

特征标群是一组写在 `(...)` 中的子模式。`(...)` 中包含的内容将会被看成一个整体，和数学中小括号 `()` 的作用相同。例如，表达式 `(ab)*` 匹配连续出现 0 或更多个 `ab`。如果没有使用 `(...)`，那么表达式 `ab*` 将匹配连续出现 0 或更多个 `b`。再比如之前说的 `{}` 是用来表示前面一个字符出现指定次数。但如果在 `{}` 前加上特征标群 `(...)` 则表示整个标群内的字符重复 `N` 次。

我们还可以在 `()` 中用或字符 `|` 表示或。例如，`(c|g|p)ar` 匹配 `car` 或 `gar` 或 `par`。

```
/(a|r)ry
```

```
grep (a|r)ry <filename>
```

图 18

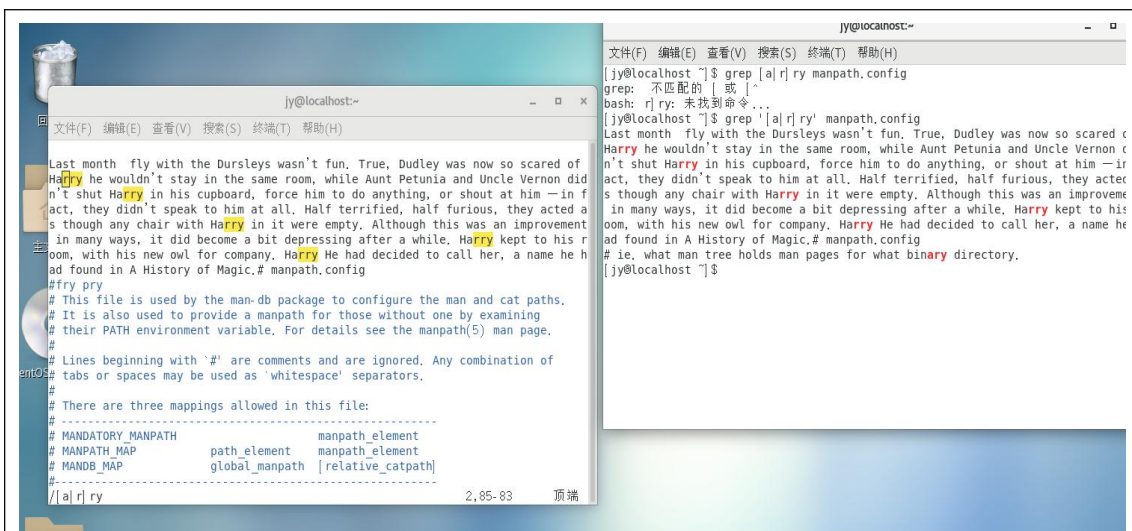


图 18 在命令模式下输入 `/ (a|r)ry`` 可以查找当前文档中以 “ary” 或 “rry” 结尾的字符串，其中 “a” 和 “r” 是可选的。在终端使用 `grep` 命令来查找文档中以 “ary” 或 “rry” 结尾的字符串，其中 “a” 和 “r” 是可选的。

第三部分

c 语言程序编辑、编译、调试

1) 通过 vim 编写如下代码:

```
#include<stdio.h>

int main()
{
    int x = 0;
    int sum = 0;
    for(x=1;x<100;x=x+2)
    {
        sum = sum+x;
    }
    printf(“sum=%d\n”,sum);
    return sum;
}
```

图 19

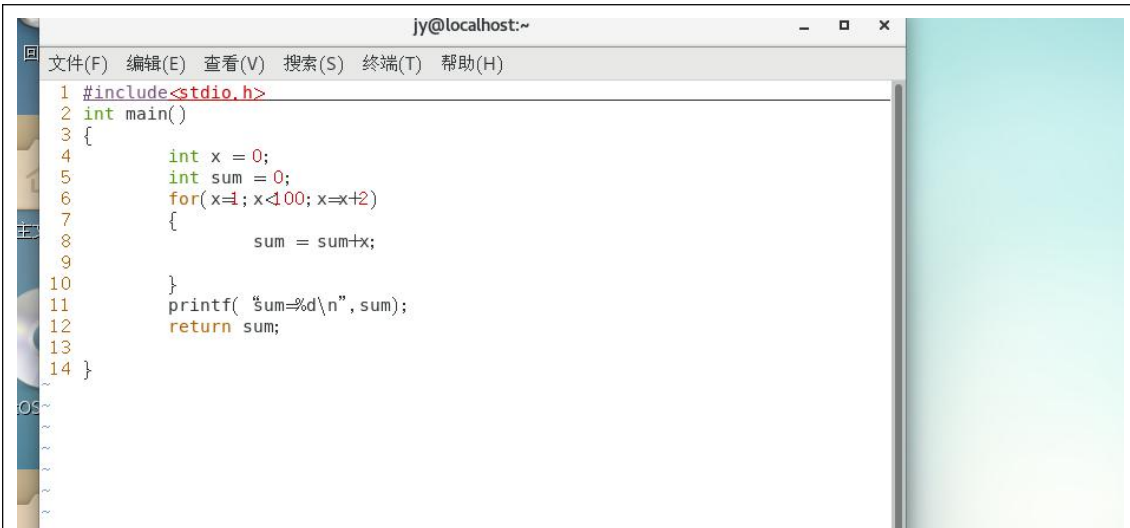


图 19 vim 编写了一个 C 语言程序

2) 使用 gcc 对上述代码进行预编译、编译、汇编、连接，最后执行该代码，五个过程的指令以及最终执行结果需要截图

预编译

图 20

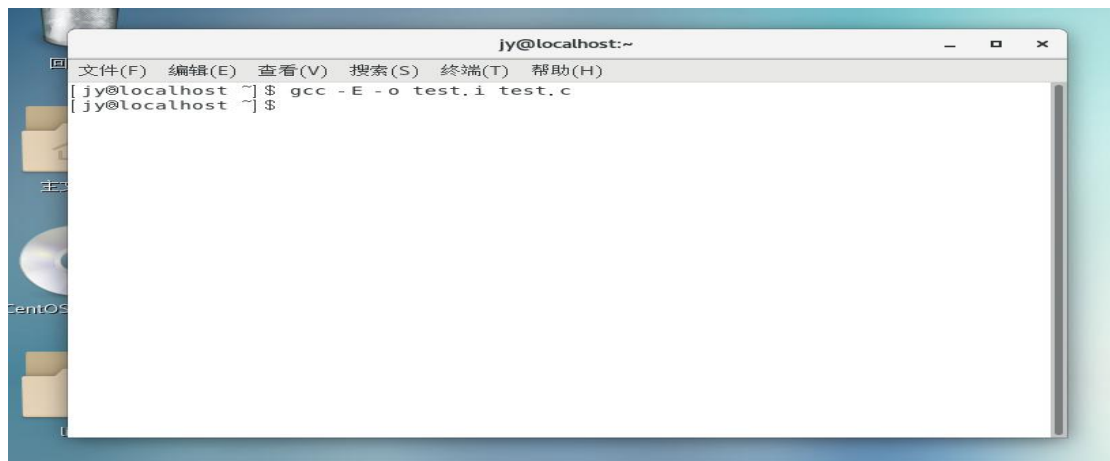


图 20 将源代码文件 test.c 进行预处理操作，生成预处理后的代码文件 test.i

编译

图 21

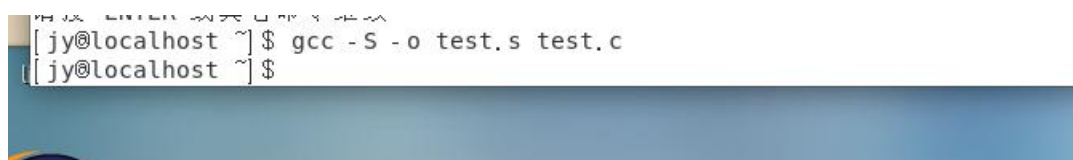


图 21 将预处理后的代码转化为汇编代码

汇编

图 22

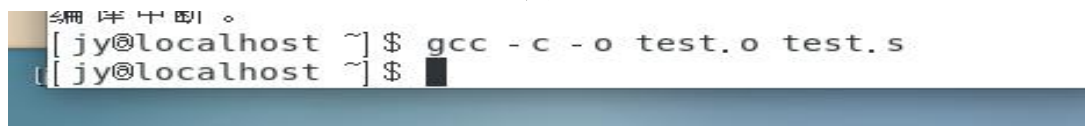


图 22 将 test.s 文件汇编为机器码文件 test.o
连接

图 23

```
[jy@localhost ~]$ gcc -o test test.o  
[jy@localhost ~]$
```

图 23 将 test.o 目标文件与必要的库函数进行链接生成可执行程序 test
执行

图 24

```
[jy@localhost ~]$ gcc -o test test.o  
[jy@localhost ~]$ ./test  
sum=2500  
lin[jy@localhost ~]$
```

图 24 运行名为“test”的可执行程序

3) 请使用尽可能多的算法编程实现 a=3, b=6 两个数字交换, 并在 linux 实现
执行

Java

图 25

```
jy@jy-virtual-machine:~/桌面$ java man.java  
交换前, a = 3, b = 6  
交换后, a = 6, b = 3  
jy@jy-virtual-machine:~/桌面$
```

图 25 使用 java 语言编写了两个数字交换的程序

C 语言

图 26

```
jy@localhost:~/桌面  
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)  
[jy@localhost 桌面]$ gcc -o c c.c  
[jy@localhost 桌面]$ ./c  
交换前, a = 3, b = 6  
交换后, a = 6, b = 3  
[jy@localhost 桌面]$
```

图 26 使用 c 语言编写了两个数字交换的程序

Python

图 27

```
netto
[root@localhost 桌面]# python man.py
('a =, 6)
('b =, 3)
[root@localhost 桌面]#
```

图 27 使用 Python 语言编写了两个数字交换的程序

C++

图 28

```
立 文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
4 [ jy@localhost 桌面] $ g++ cpp.cpp -o cpp
  [ jy@localhost 桌面] $ ./cpp
  a = 6
  b = 3
  [ jy@localhost 桌面] $
```

图 28 使用 c++语言编写了两个数字交换的程序

第四部分

完成 vim 的自定义化配置

在家目录中创建一个 `~/.vimrc` 文件，如需安装插件，在 `加` 目录下创建 `.vim`，并在其下创建 `bundle` 目录，插件装在里面。（需安装 `Vundle` 插件管理器）。

下载 Vim 插件

1.使用 `touch` 和 `mkdir` 创建目录和文件

```
touch ~/.vimrc
```

```
mkdir -p ~/.vim/bundle
```

图 29

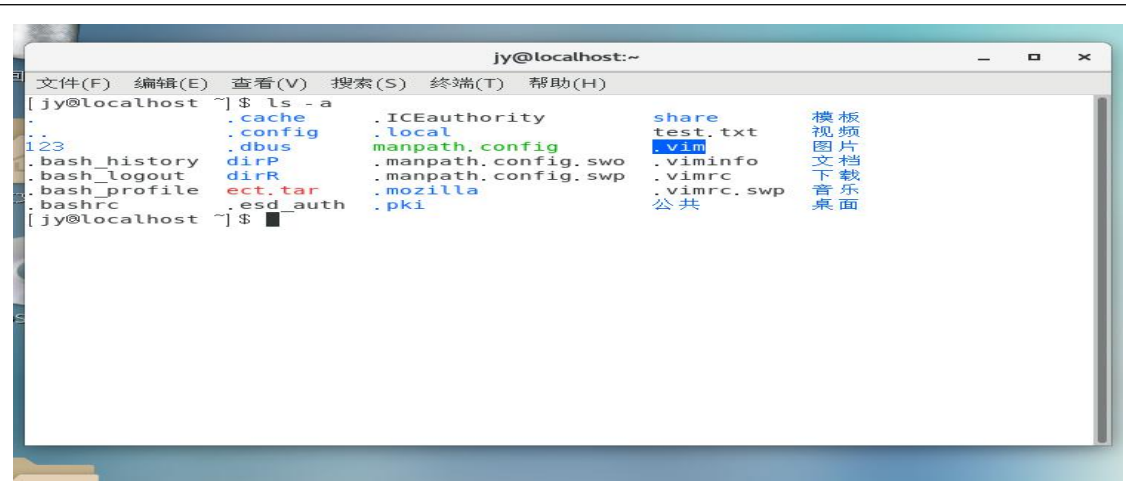


图 29 在当前目录使用 `touch ~/.vimrc` , `mkdir -p ~/.vim/bundle` 创建了一个隐藏的文件和文件夹

2.安装 vundle

```
git clone https://github.com/VundleVim/Vundle.vim.git ~/.vim/bundle/Vundle.vim
```

图 30

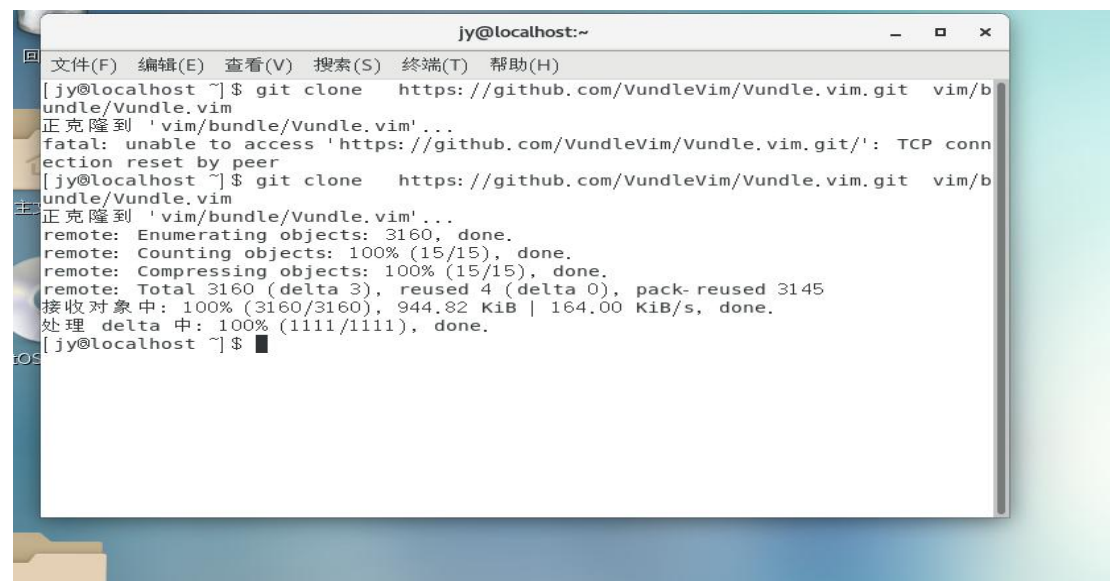


图 30 将 github 上的 vundle.vim 仓库克隆到了自己创建的隐藏文件夹中

3.对 vim 的配置文件进行配置

```
vim ~/.vimrc
```

图 31

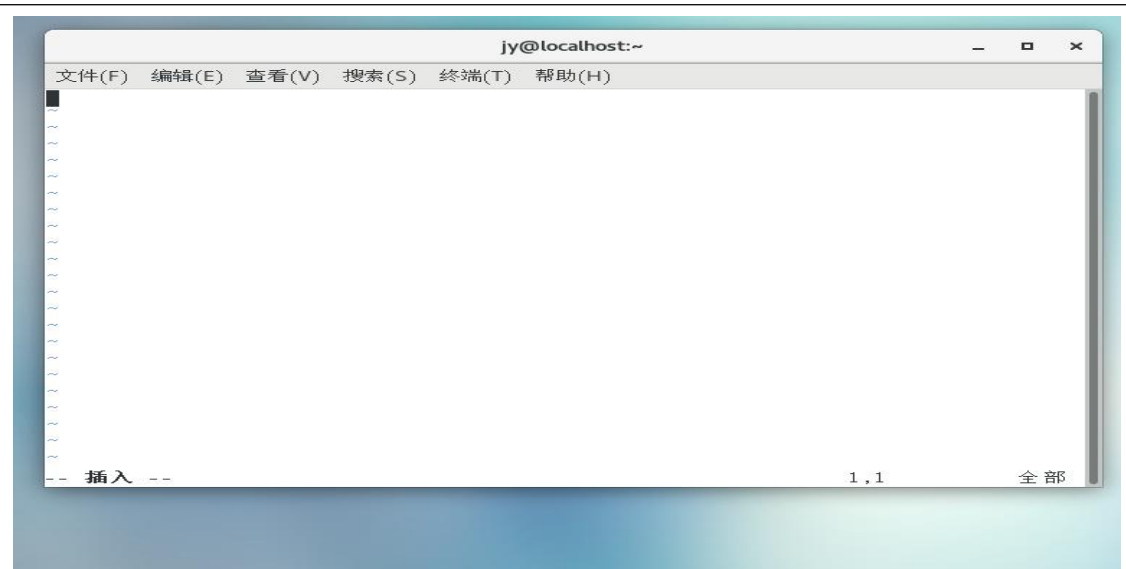


图 31 使用 `vim ~/.vimrc` 命令来配置 `.vimrc` 文件

4.配置内容可参考如下内容，请选择 10-15 项进行填写配置

"缩进量为 4 空格

`set tabstop=4`

`set shiftwidth=4`

"禁用 vi 兼容模式

`set nocompatible`

"vim 的内部编码

`set encoding=utf-8`

"vim 在与屏幕和键盘交互师傅使用的编码（取决于实际的终端的设定）

`set termencoding=utf-8`

"vim 打开文件时的尝试使用的编码

`set fileencodings=utf-8,gbk,gb2312,default,latin1`

"开启语法高亮功能

`syntax enable`

"允许用指定高亮替换方案

`syntax on`

"显示行号

`set nu`

"括号匹配

`set showmatch`

"自动检测文件类型

`filetype plugin indent on`

"在缩进和遇到 Tab 键时使用空格替代

`set expandtab`

"根据文件类型设置缩进格式

```
au FileType html,vim,javascript setl shiftwidth=2
au FileType html,vim,javascript setl tabstop=2
au FileType java,php,pathon setl shiftwidth=4
au FileType java,php,pathon setl tabstop=4
"启动 vim 时不要自动折叠代码
set foldlevel=100
"自动对齐
set ai
"根据下面的格式对齐
set si
set smarttab
set wrap
set lbr
set tw=0
set foldmethod=syntax
"总是显示状态栏
set laststatus=2
"显示光标当前位置
set ruler
"高亮显示当前行
set cursorline
"高亮显示搜索结果
set hlsearch
"禁止拆行
set nowrap
"设置快捷键前缀，即<Leader>
let mapleader=":"
"设置 tags 文件为当前目录下的 tags，默认情况也是
set tags+=./tags
"配置文件.vimrc 更改后自动重新载入使设置生效
autocmd! bufwritepost .vimrc source ~/.vimrc
"vim 对一些流行的编程语言的语法提供自动完成的功能，但是默认是没有
启用。
autocmd FileType php set omnifunc=phpcomplete#CompletePHP
autocmd FileType html set omnifunc=htmlcomplete#CompleteTags
autocmd FileType css set omnifunc=csscomplete#CompleteCSS
autocmd FileType java setlocal omnifunc=javacomplete#Complete
```

图 32

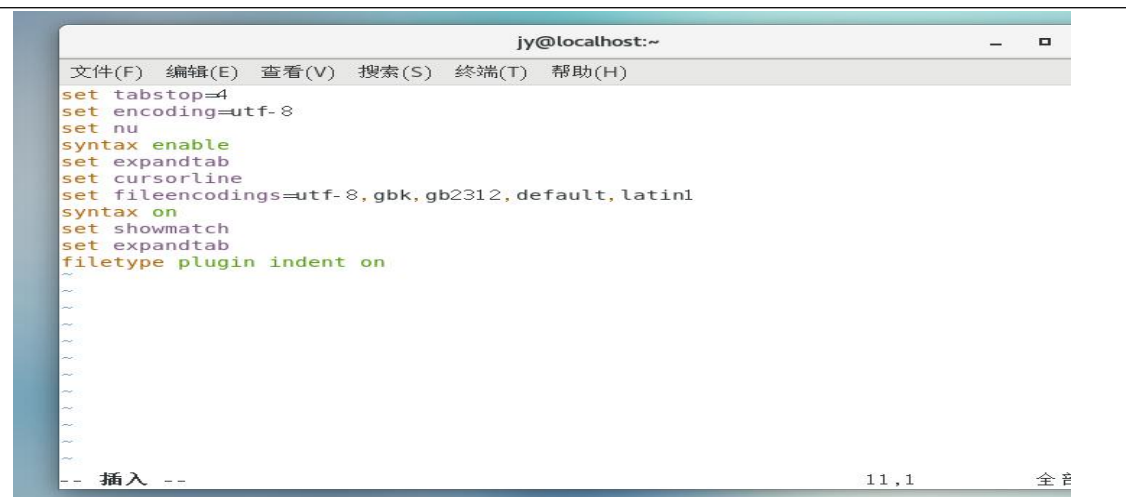


图 32 在.vimrc 文件中填写需要的配置

5.对 vundle 插件管理器进行配置，并增加插件

" 设置包括 vundle 和初始化相关的 runtime path

set rtp+=~/ .vim/bundle/Vundle.vim

call vundle#begin()

" 让 vundle 管理插件版本,必须

Plugin 'VundleVim/Vundle.vim'

" 插件列表内容，插件内容必须在 begin()和 end()之间

Plugin 'scrooloose/nerdtree'

Plugin 'jistr/vim-nerdtree-tabs' " NERDTree 共享插件

Plugin 'kien/ctrlp.vim' " 文件快速搜索插件

Plugin 'taglist.vim' " ctags + taglist 插件

call vundle#end() " 必须

图 33



图 33 在 vimrc 文件中对 vundle 插件管理器进行配置，并增加插件

6. 自定义增加新的插件

图 34

```

29
30
31 Plugin 'tjake/vim-surround' "简化对文本的编辑和修改
32 Plugin 'tjake/vim-fugitive' "可以在 Vim 中快速完成 Git 操作
33 Plugin 'vim-airline/vim-airline' "提供了漂亮的状态栏和标签栏功能
34 Plugin 'preservim/nerdtree.vim' ":NERDTree提供一个侧边栏浏览文件的文件树
   目录
35 Plugin 'tjake/vim-unimpaired'
36 Plugin 'lyokha/vim-xkbswitch'
37 Plugin 'jiangmiao/auto-pairs'
38 Plugin 'godlygeek/tabular'
39 Plugin 'boydos/emmet-vim'
40 Plugin 'jszakmeister/markdown2ctags'
41 Plugin 'junegunn/fzf.vim'
42 Plugin 'tjake/vim-commentary'
43 Plugin 'ycm-core/YouCompleteMe'
44 Plugin 'ludovicchabant/vim-gutentags'
45 Plugin 'ascenator/L9'
46 "更新插件 BundleUpdate
47 "列出所有插件 BundleList
48 "查找插件 BundleSearch
49 call vundle#end() " 必须
NORMAL .vimrc vim utf-8[unix] 100% ln: 49/49 361

```

图 34 在 vimrc 文件中增加了自己想要的插件

四、实验总结

通过本次实验，我深入了解了 Linux 系统中的文本处理技术和命令，掌握了它们的基本用法和特点同时，我也意识到 Linux 文本处理在实际工作和学习中的重要性。

Vim 有两种模式，命令模式和编辑模式。命令模式用于执行命令和移动光标，而编辑模式用于输入和编辑文本。了解和掌握这两种模式的切换和使用方法是 Vim 的基本功。通过使用冒号(:)可以进入命令行模式，执行一些高级命令和配置操作。例如，通过 :wq 命令可以保存并退出 Vim，通过 :set nu 命令可以在 Vim 中显示行号。在 Vim 中，可以使用 / 和 ? 命令来搜索和查找文本。同时，可以使用 :s 命令来进行替换操作。Vim 有很多插件可以扩展其功能。Vundle 和 Pathogen 是常用的 Vim 插件管理器，可以方便地安装和管理 Vim 插件。本次实验安装 vundle 插件管理器，也下载了几个来实验了一下。

通过这些实验，我学习并巩固了 Vim 的基本操作和高级功能。我相信这些知识和经验将在我的日常使用中帮助我更加高效地编辑和处理文本。

实验成绩评定表

序号	考核项目	分值分布	成绩
1	出勤与纪律情况	10	
2	实验任务完成情况	40	
3	实验报告质量	50	
总分			
指导教师签字			

备注：考核项目及分值分布应依据教学大纲确定。